# Will 2026 be the warmest year on record?

Climate · confidence: high · resolves YES if the Copernicus annual report (ERA5) names 2026 the warmest year or tied with 2024: an anomaly of ≥ 0.72 °C above the 1991–2020 average

TODAY · 10 NOV 2026 — NOVEMBER BULLETIN · ≈ 5 DEC — CHECK · 15 JAN 2027

**13.92%**

CONCLUSION

A new temperature record in 2026 is unlikely — 13.92%. Over ten months the year has been 0.64 °C warmer than the 1991–2020 average, 0.07 °C below the record year 2024. To match it, November and December must average 1.09 °C above normal — 0.16 °C more than the warmest month in the entire ERA5 record. October, the warmest October in the series, came in at 0.90 °C. El Niño is already at record strength and still strengthening into December, but surface temperature responds to it with a lag of 2–4 months, so the main effect will fall in early 2027. If November comes in at 1.05 °C or higher, the chance of a record rises to 57.39%.

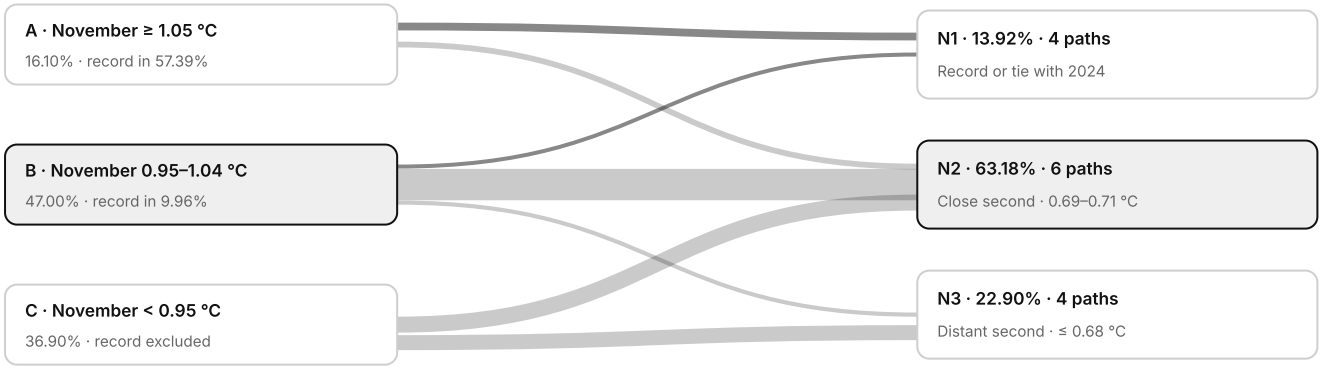
SCENARIOS · LEVELS 1–2 · CHANCE OF A RECORD IN EACH

| | | P | RECORD |
|---|---|---|---|
| E1 · strengthens — El Niño · 0.70 | A · November ≥ 1.05 °C | 0.20 | 60.00% |
| | B · November 0.95–1.04 °C | 0.50 | 12.00% |
| | C · November < 0.95 °C | 0.30 | excl. |
| E2 · plateau/decline — El Niño · 0.30 | A · November ≥ 1.05 °C | 0.07 | 40.00% |
| | B · November 0.95–1.04 °C | 0.40 | 4.00% |
| | C · November < 0.95 °C | 0.53 | excl. |

REVISION SIGNALS

- ▲ November in the C3S bulletin ≥ 1.05 °C
- ▲ C3S Climate Pulse daily data above 1.0 °C in late November
- ▲ Niño 3.4 still rising in November
- ▼ November < 0.95 °C
- ▼ CPC or BoM: El Niño has peaked
- ▼ December cooler than November

## Scenario flows




## Analytical position

The analysis proceeds from **N2 — "a close second" (63.18%)**: 2026 will fall short of 2024 by 0.01–0.03 °C. Both El Niño scenarios and every November band lead to it, because a record needs two consecutive months of unprecedented anomalies, while a wider gap needs cooling during a record El Niño. The position changes if November exceeds 1.05 °C: then **N1 — a record** becomes most likely (57.39% in that band). **N3** (22.90%) — a distant second — becomes the baseline if November stays below 0.95 °C and El Niño peaks before December. Third place is excluded: it would require November and December to cool to 0.40 °C.

## COEFFICIENTS

| Lvl | Variable | Coefficients | Why · research |
|---|---|---|---|
| 1 | El Niño trajectory to December | 0.70 · 0.30 | October: Niño 3.4 at +3.21 °C — a record anomaly, above the peaks of 1982–83, 1997–98 and 2015–16 (C3S). Copernicus expects the peak in December; CPC expects strengthening through year-end and an 83% chance of historic strength in October–December. Hence the weight of E1. |
| 2 | November anomaly | E1: 0.20 · 0.50 · 0.30<br>E2: 0.07 · 0.40 · 0.53 | September 0.82 °C → October 0.90 °C: +0.08 in a month. Keeping that pace gives ≈ 0.98 °C — band B. Band A requires twice the rate of increase and beating the ERA5 monthly record (0.93 °C, September 2023). Global temperature lags Niño 3.4 by 2–4 months (Foster, Rahmstorf). |
| 3 | December → node | A: 0.60 · 0.40 (E1)<br>B: 0.12 · 0.80 · 0.08 (E1) | Annual-mean arithmetic: a record needs November + December ≥ 2.18 °C. After band C, December would have to reach ≈ 1.28 °C — a record is excluded. After band A, a distant second is excluded: December would have to fall below 0.74 °C. |

## RESEARCH

| | |
|---|---|
| Human-induced warming of 0.27 °C per decade; 2025 at 1.39 °C above pre-industrial (Forster et al. 2026) | baseline · each year starts higher |
| The 2023 spike was driven by ENSO: after a prolonged La Niña such jumps occur in 10.3% of model cases versus 1.6% overall (Raghuraman et al. 2024) | level 1 · weight of El Niño |
| Most of the ≈ 0.2 °C of the 2023 anomaly not explained by the trend and El Niño is attributed to record-low albedo — fewer low clouds (Goessling et al. 2025) | level 2 · why band A is possible |
| Surface temperature responds to ENSO with a lag of 2–4 months (Foster, Rahmstorf 2011) | levels 2–3 · the December peak acts on 2027 |

## SOURCES


| | |
|---|---|
| Copernicus C3S. Surface air temperature for October 2026. Climate Bulletin, 06.11.2026 | **A1** |
| NOAA Climate Prediction Center. ENSO Diagnostic Discussion, 08.10.2026 | **A1** |
| Forster et al. Indicators of Global Climate Change 2025. Earth System Science Data 18, 3889–3933, 2026 | **A2** |
| Raghuraman et al. The 2023 global warming spike was driven by the El Niño–Southern Oscillation. Atmos. Chem. Phys. 24, 11275–11283, 2024 | **A2** |
| Goessling, Rackow, Jung. Recent global temperature surge intensified by record-low planetary albedo. Science 387, 2025 | **A2** |
| Foster, Rahmstorf. Global temperature evolution 1979–2010. Environmental Research Letters 6, 044022, 2011 | **A2** |


## ALL VARIANTS

| | | |
|---|---|---|
| E1 → A (≥ 1.05 °C) → December ≥ record threshold | 8.40% | **N1** |
| E1 → A (≥ 1.05 °C) → December in between | 5.60% | **N2** |
| E1 → B (0.95–1.04 °C) → December ≥ record threshold | 4.20% | **N1** |
| E1 → B (0.95–1.04 °C) → December below November | 2.80% | **N3** |
| E1 → B (0.95–1.04 °C) → December in between | 28.00% | **N2** |
| E1 → C (< 0.95 °C) → December ≥ 0.92 °C | 12.60% | **N2** |
| E1 → C (< 0.95 °C) → December < 0.92 °C | 8.40% | **N3** |
| E2 → A (≥ 1.05 °C) → December ≥ record threshold | 0.84% | **N1** |
| E2 → A (≥ 1.05 °C) → December in between | 1.26% | **N2** |
| E2 → B (0.95–1.04 °C) → December ≥ record threshold | 0.48% | **N1** |
| E2 → B (0.95–1.04 °C) → December below November | 2.16% | **N3** |
| E2 → B (0.95–1.04 °C) → December in between | 9.36% | **N2** |
| E2 → C (< 0.95 °C) → December ≥ 0.92 °C | 6.36% | **N2** |
| E2 → C (< 0.95 °C) → December < 0.92 °C | 9.54% | **N3** |

## WHAT CHANGES THE ESTIMATE

| | |
|---|---|
| November ≥ 1.05 °C | **57.39%** |
| November 0.95–1.04 °C | **9.96%** |
| November < 0.95 °C | **0%** |
| El Niño has already peaked (E2) | **4.40%** |

Limitations: the January–October average is published to 0.01 °C, so the record threshold is known only within ± 0.025 °C. The final ranking is set by the C3S annual report; the estimate does not consider other datasets (NASA, NOAA, Met Office).

Next review: the C3S bulletin for November (≈ 5 December).